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PR-FCNN: a data-driven hybrid approach for predicting PM2.5 concentration

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Abstract

The atmosphere's fine articulate Matter (PM2.5) poses various health-related risks. Even though multiple efforts have been made to lower the emissions of these substances, the mortality rate is continuously increasing, requiring immediate inclination of the scientific community towards the design and development of advanced predictive models. Conventional statistical approaches have become dormant due to their limitations in capturing the innate relationships between the pollutants, particularly for predicting PM2.5 concentrations. In contrast, machine and deep learning techniques have shown great potential for forecasting air quality, providing more accuracy than its predecessor techniques. The present study investigates the utilization of hybrid approaches by integrating machine learning models with deep learning models to improve the prediction capabilities of PM2.5 concentration. It uses datasets from the World Air Quality Index (WAQI) and the State of Global Air (SOGA) to analyze the performance of the models on both the daily and annual data, respectively. This ensures the model's effectiveness on a diversified dataset. The present study implements Random Forest (RF), Polynomial Regression (PR), XGBoost, and Extra Tree Regressor (ETR) coupled with Fully Connected Neural Network (FCNN), Long Short-Term Memory (LSTM), and Bi-directional LSTM (Bi-LSTM) for obtaining optimized results. Finally, after a thorough investigation, the hybrid PR model coupled with FCNN (PR-FCNN) is found to be the best model with improved R-squared (R^2) values, portraying its potential for predicting PM2.5 concentration accurately. Based on the experimentation, the preset study recommends implementing hybrid approaches, offering better predictive accuracy in forecasting air pollutants, especially PM2.5.

Keywords Air pollution · Neural network · PM2.5 · Human health · R-squared

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1 Introduction

The rapid increase in industrialization, extensive usage of energy, exponential growth in the population, urbanization, and warehousing and goods movement have made air pollution a global challenge impacting not only human health but also the social and economic systems [1–3]. Despite a decrease in the emission rate of air pollution in the last decade [4], the mortality rate due to air pollution is considerably increasing with it being more than the combined rate of HIV/AIDS, tuberculosis, and malaria [5]. Several studies have linked air pollution with respiratory disease [6], few showed its association with mental health [7], some examined its association with the risk of cancer [8], some linked it with the risk of psoriasis [9], while others have linked it with corporate investments and policy-making [10, 11], enforcing the challenges with the quantification and modeling of air pollution [6]. Numerous approaches have been proposed to study this impact, including Land Use Regression (LUR), remote sensing inversion, interpolation methods, and deterministic modeling [6].

Most importantly, the scientific community is focused on identifying and understanding how air pollution contributes to human health problems [12]. In humans, children are more vulnerable to air pollutants, as their lungs are not fully developed, which maximizes the chances of exposure than adults as different chemical compositions can remain in the lungs for a very long time [13]. People with diabetes, heart diseases, and atopic patients are sensitive to air toxicants. [14]. Inclusive of humans, some studies have also investigated the effects of air pollutants on the physiological processes in plants [15]. Both anthropogenic and physical activities are responsible for the spread of different pollutants in the environment [16].

Air Pollution is mainly caused by the significant release of greenhouse gases, biological matter, and particulate matter into the atmosphere [17]. The only way to handle the adverse impact of air pollution is by identifying the patterns and inherent associations and predicting the effect in advance, as air pollution heavily relies on the spatial and temporal correlations of prior air pollution structures [18]. The increase in air pollutants will also affect the overall climate [19]. Based on the physical and chemical properties, air pollutants are classified into criteria and toxic air pollutants, where the former includes particulate matter with a diameter of 2.5 µm and 10 µm, Lead, Nitrogen dioxide, Ozone, Carbon Dioxide, and Sulfur Dioxide, in contrast, the latter includes 189 other air pollutants [20]. The leading causes of poor air quality are the pollutants of particulate matter of micro sizes, oxides of Nitrogen (NO_x), Sulphur Dioxide (SO_2), Carbon Monoxide, Ozone, and Volatile Organic Carbons (VOCs) [2]. Due to its high effect on human health, PM2.5 is especially a pollutant among others [2]. It is pertinent today to accurately predict air quality with higher accuracy to prevent harmful medical impacts caused by air pollution [21]. In recent decades, key air pollutants were predicted using deterministic and statistical methods [21]. The particulate matter (PM2.5) is mainly of serious concern as its increase in the air can cause severe health conditions [22]. As per the report of the World Health Organization (WHO), the air breathed by approximately 90% of people all over the globe is harmful and does not comply with the guidelines of WHO Air Quality Guidelines [23]. PM2.5 is considered extremely dangerous due to its capability to penetrate the respiratory system of humans [23]. The impact of particulate matter, especially PM2.5, enforces the need for research into its toxic effects and chemical compositions [24]. PM2.5 refers to the particulate matter with a diameter < 2.5 µm with aerodynamic characteristics to remain in the atmosphere longer [24]. Due to the petite size, heavy biological activities, and large surface area, the surface of PM2.5 is attacked by several chemical components that play critical roles in human and animal health, particularly during the growth and development stages [24]. Although the concentration of sulfur dioxide, nitrogen dioxide, and ozone have considerably decreased over a decade, researchers' attention has now shifted to particulate matter, especially with a diameter of less than 2.5 µm [25].

The rest of the article is divided into the following sections. In Sect. 1.1, we have examined the previous state-of-the-art methods and the evaluation metrics for predicting the PM2.5 concentration in the air. Afterward, in Sect. 2 section, we first discussed the data collection and datasets used in the present study, along with data pre-processing and model selection procedures. Furthermore, in Sect. 3, we thoroughly investigated different hybrid approaches and evaluated their performances to select the best-performing model. In Sect. 4, we summarize the study's findings and recommend further directions.

1.1 Background

Machine and deep learning have been widely incorporated in research, from image processing [26, 27], fuel simulation [28, 29], and atmospheric research [21], and they are massively employed in forecasting air pollutants such as

PM2.5 concentrations [24]. Recent studies have indicated the effectiveness of machine learning techniques [30–32] over traditional statistical methods [33, 34]. Presently, considerable use of deep learning algorithms to forecast the concentration of PM2.5 is seen, mainly due to the recent advancement in deep learning approaches [31]. Many studies have explored the vast application of machine and deep learning techniques for forecasting and estimating the trends of various key air pollutants, emphasizing these techniques' impact on producing accurate and efficient results. The researchers [17] compared the performance of various machines and deep learning to predict air quality. They found XGBoost to be the best-performing algorithm for predicting the concentration of air pollutants [17]. Muthukumar et al. integrated Convolutional LSTM (ConvLSTM) and Graph Convolutional Network (GCN) architecture to predict PM2.5 by using multi-source image and sensor data [18], whereas, in [35] Muthukumar et al. again utilized a similar coupled architecture for predicting PM2.5 concentration by using big remote-sensing data. Additionally, Pak et al. implemented a hybrid approach CNN-LSTM model to predict the concentration of PM2.5 [21], and the study by Shakya et al. proposed the Gated Recurrent Unit Based Encoder-Decoder (GRU-ED) method prediction of PM2.5 [22]. Furthermore, the researchers [23] in their study proposed a Weight Long-Short Term Memory Neural Network Extended (WLSTME) to handle the effect of location density as well as wind conditions on the concentration of air pollution on the spatiotemporal correlation. Similarly, the researchers in [36] developed a deep learning model, EntityDenseNet, for extracting the ground-level real-time concentration of PM2.5 in the air [36]. Researchers in [37] designed a coupled CNN-LSTM model with spatiotemporal features by integrating historical and meteorological data of air pollutants collected at adjacent stations [37]. Researchers in [38] suggested using LSTM as the best model for predicting PM2.5 concentration out of the investigated ARIMA, FBProphet, and 1D-CNN. Also, the researchers in [39] proposed a hybrid LSTM and Deep Auto-Encoder (DAE) method on hourly data of PM2.5 concentration. The study conducted by researchers in [40] proposed a hybrid approach of LSTM-FCNN for PM2.5 concentration to handle long-range dependencies [40]. Also, the researchers in [41] proposed an ensemble of Deep Neural Network-based regression for image data utilizing a Feedforward Neural Network to obtain combined PM2.5 prediction by coupled CNN consisting of VGG-16, Inception-V3, and ResNet50 [41]. The researchers in [42] proposed a ConvLSTM model to manipulate the spatial and temporal features of the data automatically [42]. In [43], the authors proposed a combination of CNN and LSTM to predict the concentration of PM2.5, and in [44] they investigated Multiple Additive Regressive Trees (MART), a Deep Feedforward Neural Network (DFNN), and an LSTM-based hybrid deep learning model and suggested LSTM as the best-performing model for predicting PM2.5 concentration [44]. The study in [45] examined the performance of a Support Vector Machine (SVM), Random Forest (RF), LSTM, and a Multi-Layer Feedforward Neural Network (MLFFNN) and suggested LSTM as the best-performing model [45] In contrast, a similar type of study was conducted by researchers in [46] evaluated several models and suggested XGBoost as the best-performing model [46]. In [19], the authors designed a multimodal deep neural network (M^2-APNET) coupled with Bi-LSTM by utilizing satellite images to predict air pollutants at the global level [19]. Also [47] after thorough evaluation, the study in [47] recommended XGBoost as the best-performing model. Additionally, the researchers in [48] investigated several algorithms and a couple and proposed a coupled approach of Adaptive Boosting of Long-Short Term Memory (AdaBoost-LSTM) models along with kernel principal component analysis (KPCA) for ensemble learning to predict the concentration of PM2.5 [48]. In [49], the researchers proposed a Temperature-based Deep Belief Network (TDBN) for predicting the daily concentration of PM2.5 [49] and in [50] the researchers resolved the structure parameter issue of the Deep Belief Network (DBN) causing a significant impact on the prediction of computational time and accuracy of PM2.5 concentration by optimizing the DBN with the introduction of Modified Grey Wolf Optimization (MGWO) algorithm [50]. Mahajan et al. proposed an empirical approach using the Neural Network Autoregression (NNAR) method for predicting PM2.5 hourly [51] In contrast, Lv et al. proposed an empirical approach using non-linear regression models to forecast daily PM2.5 levels [52]. Also, the study conducted by researchers in [53] proposed a hybrid model by integrating Ensemble Empirical Mode Decomposition (EEMD) and General Regression Neural Network (GRNN) based on the preprocessing and analysis of the data for the prediction of the concentration of PM2.5 in the air [53]. Considering the variation in the diversity of datasets utilized for the prediction of PM2.5 concentration by many researchers, there is a need for a model that can perform equally encounter both the dataset of daily and hourly data and can produce an efficient and optimized prediction of the concentration of PM2.5 in the air.

2 Methodology

In the present study, we have investigated various machine learning, deep learning, and hybrid techniques to examine their performances using multiple evaluation metrics. We have also proposed a novel hybrid approach for predicting the concentration of PM2.5. We investigated by collecting data from various resources to obtain authentic datasets. We selected valid datasets of the World Air Quality Index (WAQI) and the State of Global Air (SOGA) as both datasets provide global air quality trends. We began by pre-processing the data and selected various models to examine their performance on the dataset. Finally, we discussed the evaluation metric used to examine the model's performance in the present study. The entire process of investigation is shown in the Fig. 1 below:

2.1 Data collection

2.1.1 World Air Quality Index (WAQI)

The World Air Quality Index (WAQI) is a global project with data from around 380 major cities and is updated thrice daily. This dataset aggregates the data from various sources and monitoring stations. In the present study, we have utilized the dataset of July 2024, which contains 11,50,598 records and 09 features. The features include Date, Country, City, Species, Count, Count, Count, Count, Count, Min, Max, Median, and Variance. The Species column contains different pollutants, and the present study has investigated the PM2.5 pollutant. The visual depiction of the WAQI dataset and its relevant features utilized in the present study are shown in Figs. 2, 3, 4 and 5.

2.1.2 State of global air

State of Global Air (SOGA) provides datasets for air quality analysis across the globe. The datasets are open access and are freely available, providing data related to several significant pollutants such as PM2.5, Ozone, and Nitrogen Dioxide. In the present study, we have only considered the dataset of PM2.5 pollutants, which contains 6325 records and 14 features. The features include Exposure ID, Type, Country, ISO3, Region, Name, Exposure Lower, Exposure Mean, Exposure Upper, Year, Pollutant, Pollutant Name, Region Name, and Units. The visual representation of the main features of the SOGA dataset utilized in the present study is shown in Figs. 6 and 7.

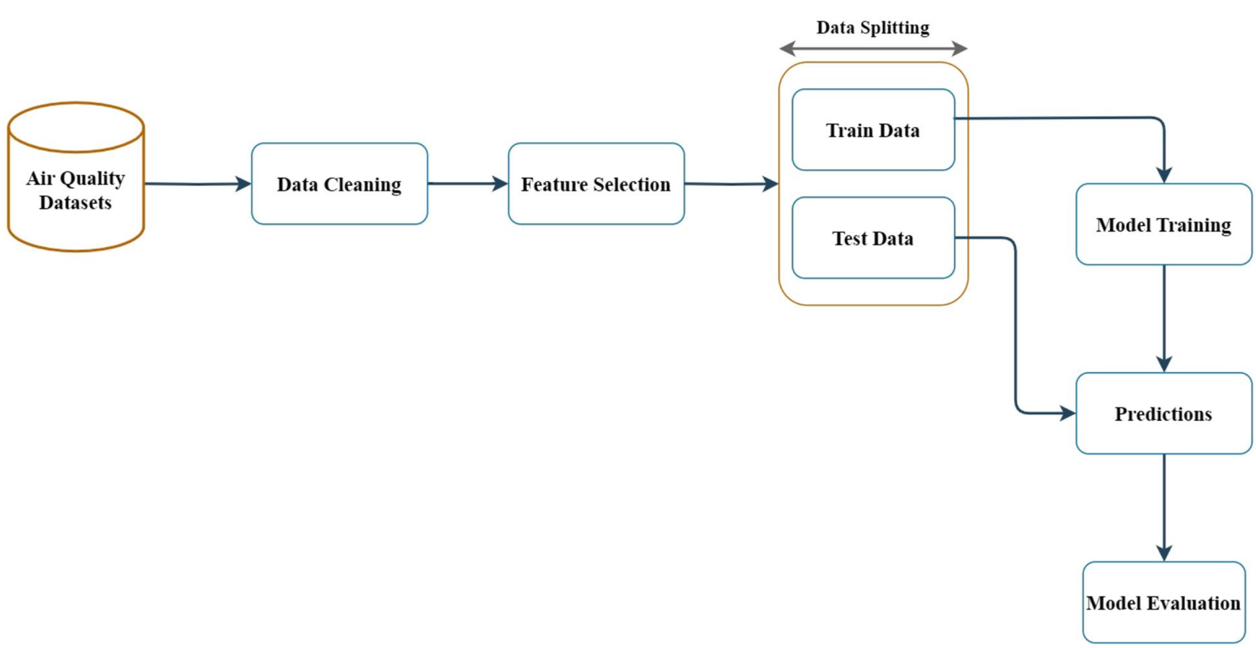

Fig. 1 From data to evaluation

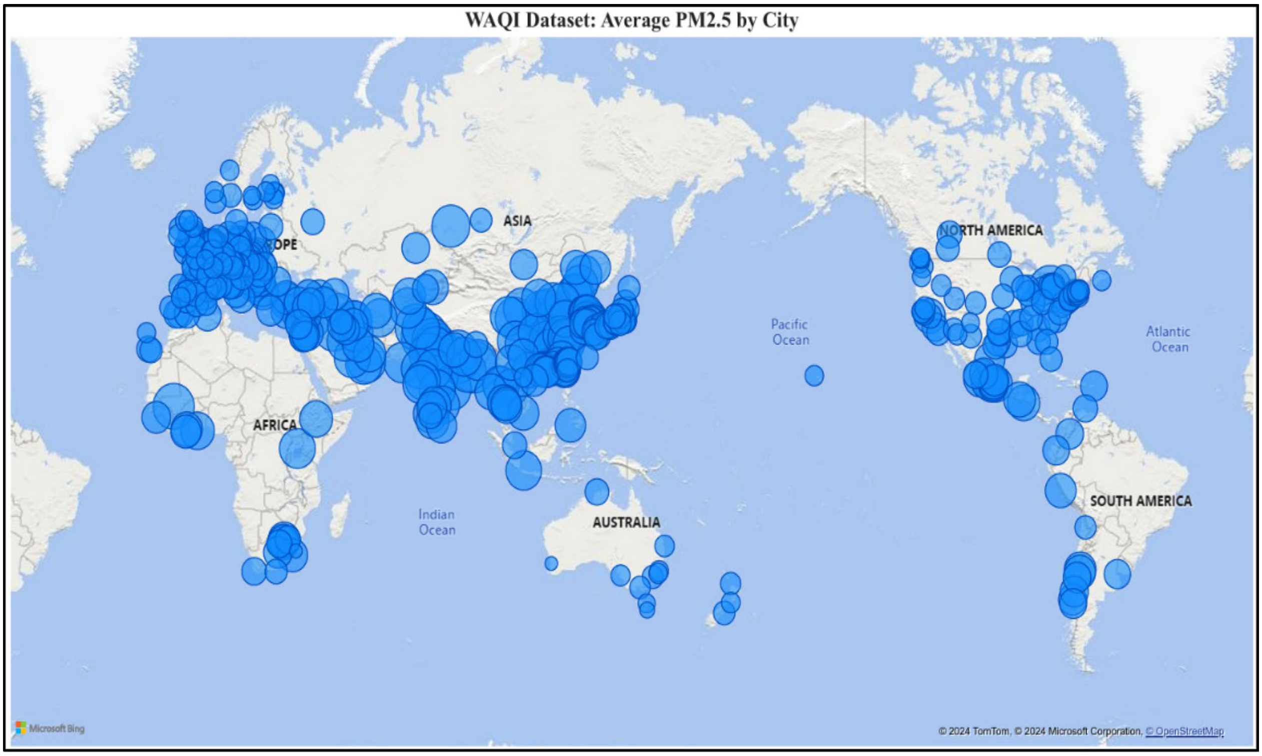

Fig. 2 Average PM2.5 concentration by city

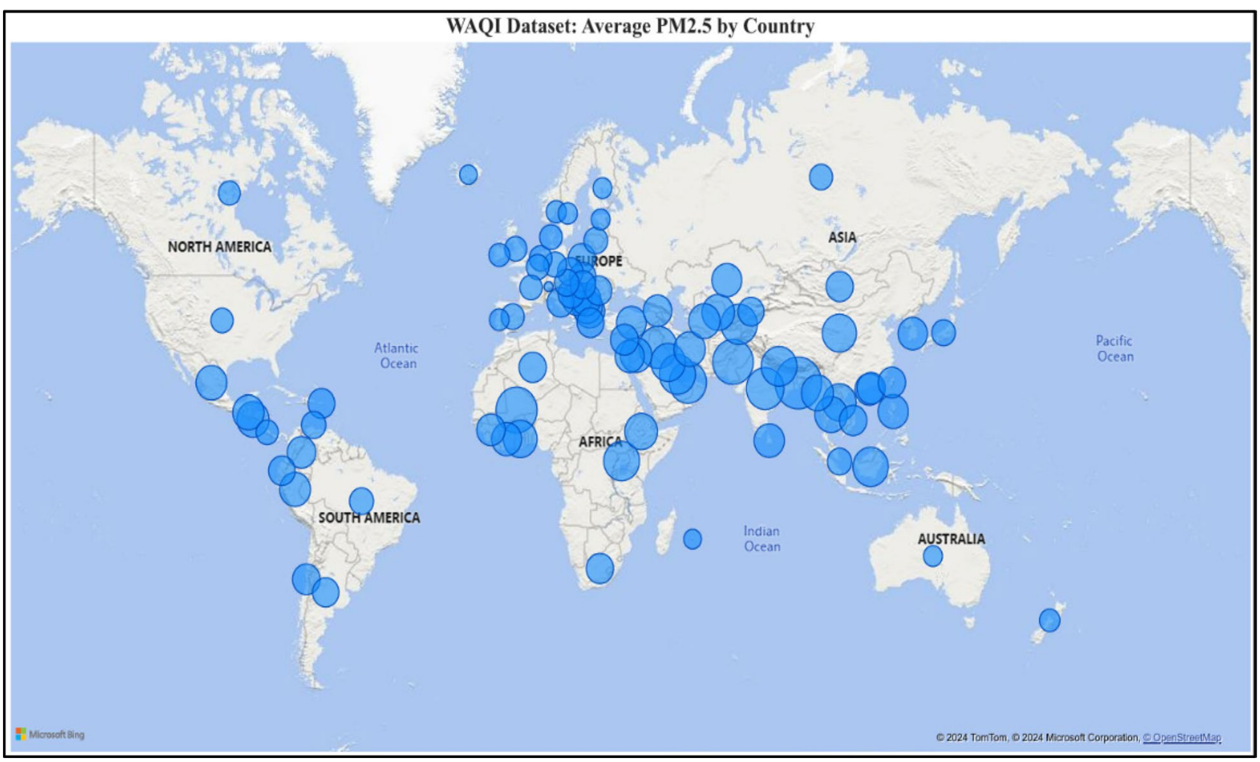

Fig. 3 Average PM2.5 concentration by country

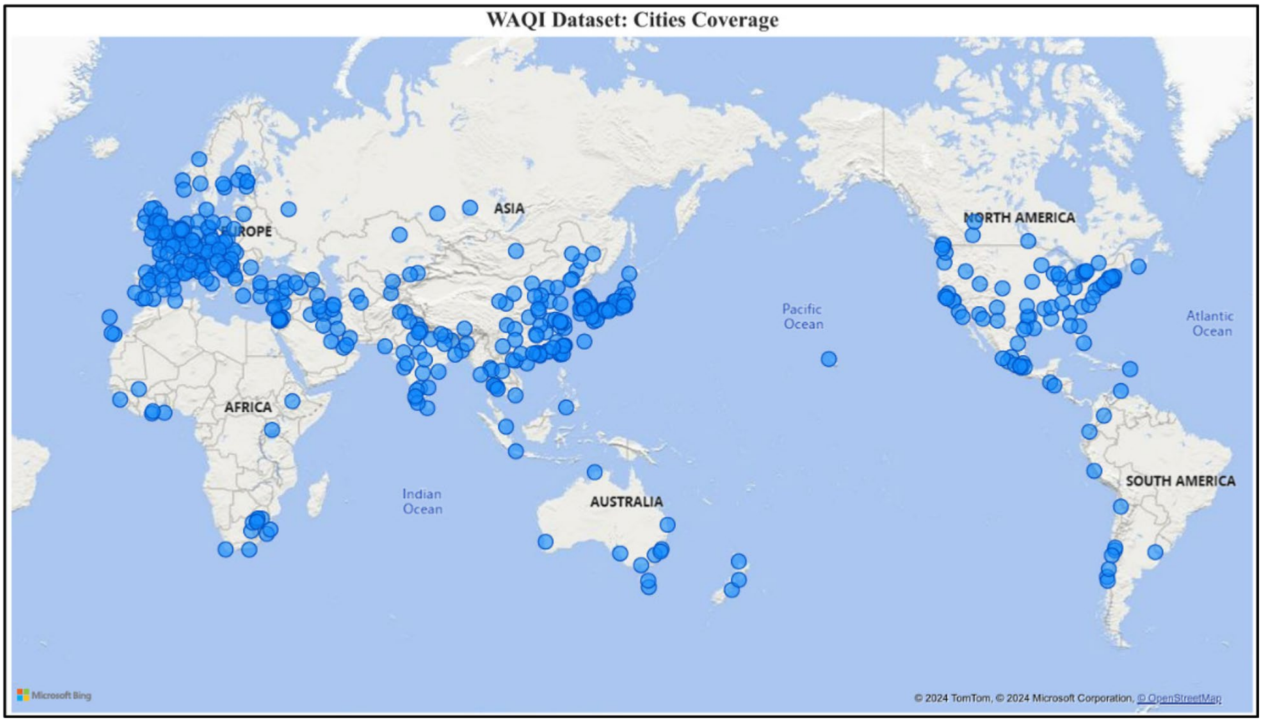

Fig. 4 Cities covered by WAQI dataset

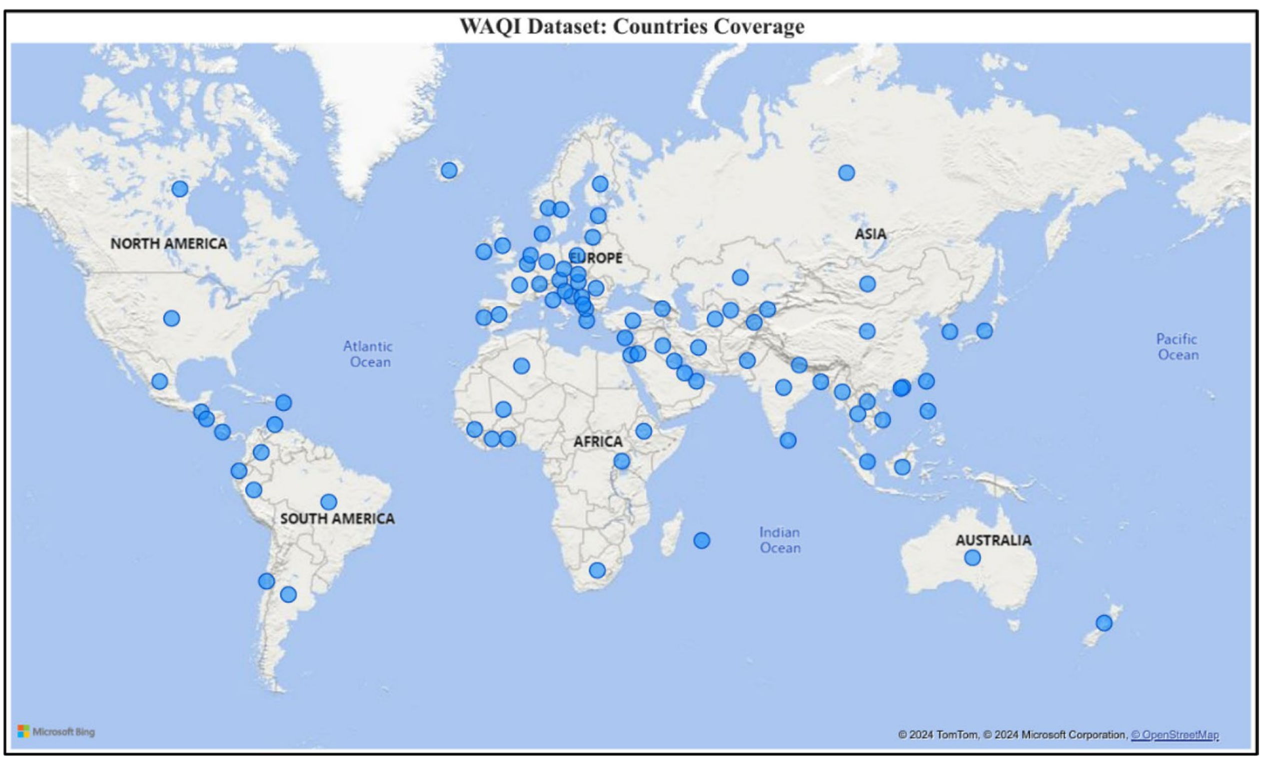

Fig. 5 Countries covered by WAQI dataset

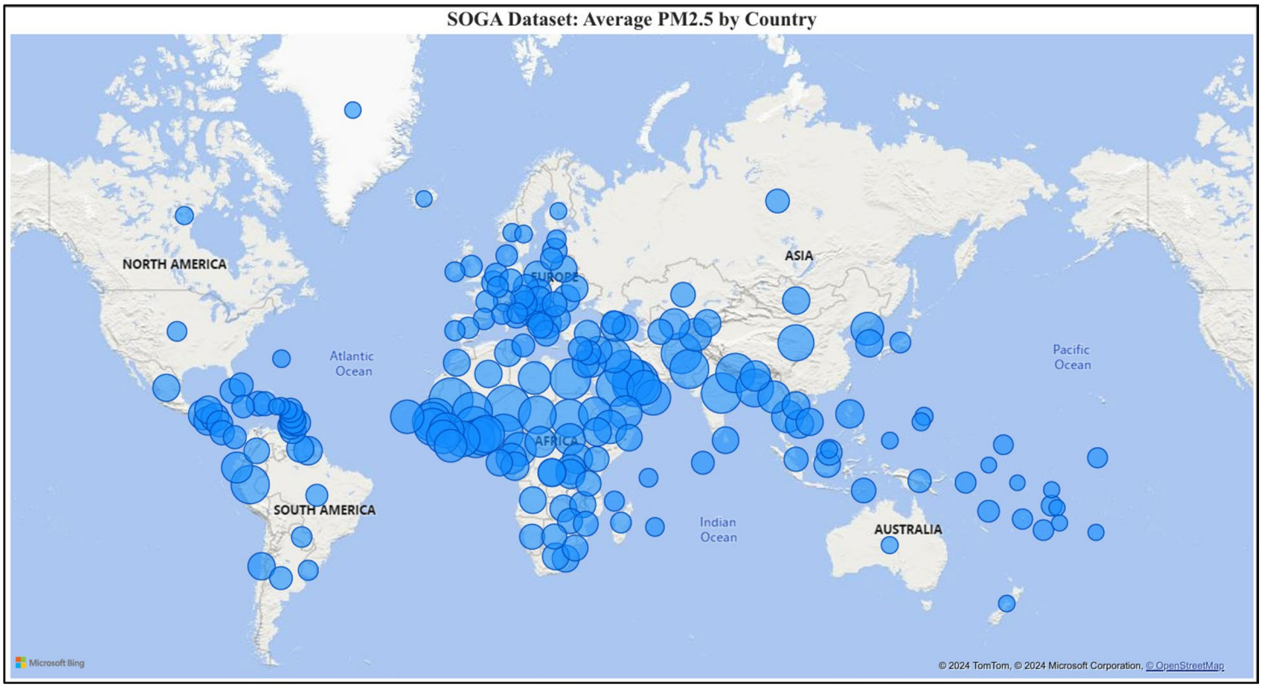

Fig. 6 Average PM2.5 concentration by country

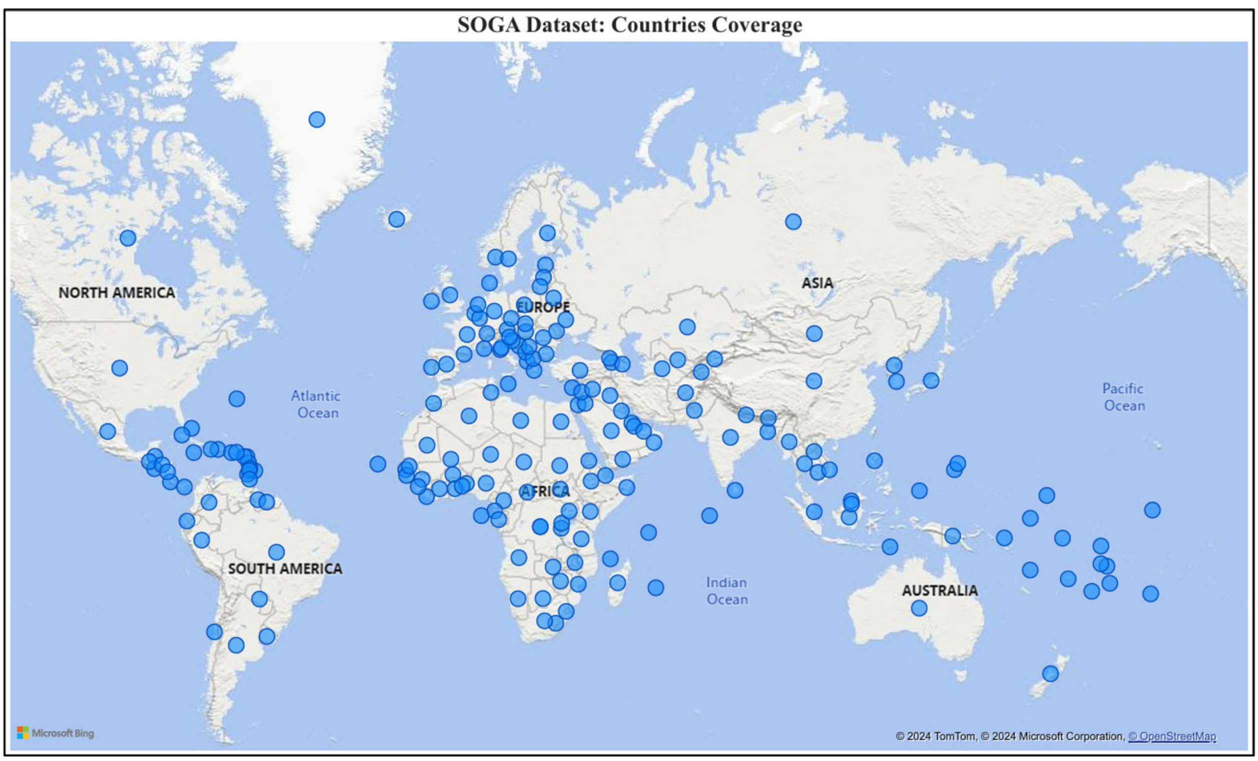

Fig. 7 Countries covered by SOGA dataset

2.2 Data preprocessing

Both the datasets used in the present study have undergone multiple pre-processing procedures to ensure that the dataset is well-structured before being incorporated into the model for evaluation. The general steps used in the present study for pre-processing the datasets are as follows:

2.2.1 Feature selection

The relevant features were identified and extracted from datasets involving dates, geographic location, and pollutant values.

2.2.2 Data filtering

The datasets were filtered to get only the records with PM2.5 pollutants. This step ensured that the analysis was focused on the contaminants of interest in this study.

2.2.3 Handling missing values

The datasets were examined for missing values, and the missing data was handled to ensure no missing values.

2.2.4 Data type conversion

To facilitate time-series analysis, date strings were converted to date time objects, and numerical features were converted to numeric data types such as int and float.

2.2.5 Normalization

Numerical features were normalized using MinMax scaling. This ensures that features are scaled to a standard range [0,1], which can improve model performance as all features contribute equally to the model's learning process.

2.2.6 Encoding

Categorical features such as geographic locations were encoded using one-hot encoding, transforming the categorical values into binary vectors for practical model training.

2.2.7 Data splitting

The preprocessed data is split into training and testing sets at an 80:20 ratio, where 80% of the data is used for training and 20% for testing. Data splitting enabled the model's performance to be evaluated on unseen data, which provides unbiased results of its predictive capabilities.

2.3 Model selection

In the present study, we have focused on utilizing the hybrid modeling approach. The machine learning models selected for the present study are linear regression, random forest regression, extra trees regression, XGBoost regression, and polynomial regression. For hybrid approaches, we integrated the machine learning models with the deep learning models of Fully Connected Neural Networks (FCNN), Long Short-Term Memory (LSTM), and Bidirectional Long Short-Term Memory (Bi-LSTM). Each machine learning model was trained for the hybrid approach, and then the predictions of that model were used as augmented features for the deep learning model. This approach allows the model to capture all the nuances of the data and improve performance by making the machine learning models capture features or patterns.

Their predictions become the additional inputs to a deep learning model that makes the final prediction. For the tools, Keras was utilized for deep learning models, and sci-kit-learn was used for the machine learning-based models except for XGBoost Regressor, which has a separate library, i.e., called XGBoost.

2.3.1 Linear regression

Linear regression is a supervised machine learning algorithm for predicting continuous values. It calculates the linear relationship between the independent features and a dependent feature by computing a linear equation on the data. It may only perform well if the relationship between the features is linear. In this study, a linear regression model is trained to capture the relationship between the features, perform predictions, and evaluate the results.

2.3.2 Random Forest Regressor (RFR)

Random Forest Regressor is an ensemble learning algorithm that constructs multiple decision trees during the training process and outputs the average prediction of the individual trees for regression-based tasks. It helps in improving generalization by training each tree on a random subset of data. It also benefits datasets with many features and can capture non-linear relationships. This study also utilizes Random Forest for comparative analysis alongside other linear and nonlinear machine learning algorithms.

2.3.3 Extra Trees Regressor (ETR)

Extra Trees Regressor, or Extremely Randomized Trees, is an ensemble learning algorithm. It is like the Random Forest; however, with few modifications in the Random Forest, the best split is chosen from a random subset of features, whereas the Extra Tree Regressor randomizes the split points within each feature. This increases randomness and leads to faster training times. It can also improve model performance by reducing the variance even further than RF. The present study uses an ETR for comparative analysis to improve performance alongside different ensemble algorithms.

2.3.4 XGBoost Regressor

XGBoost Regressor or Extreme Gradient Boosting Regressor is an advanced implementation of gradient designed for speed and performance. It constructs an ensemble of decision trees, where each tree attempts to minimize errors made by the previous trees. The final prediction is achieved by taking an average of predictions from all the trees. Its regularization techniques prevent overfitting, it can also handle missing data, and it is highly customizable. In the present study, it is utilized as a boosting algorithm.

2.3.5 Polynomial Regression (PR)

Polynomial Regression is like linear regression and associates the independent and dependent features to find the best-fit line through the data points. It adds polynomial terms of input features to the linear model, allowing the model to fit a non-linear relationship between independent and dependent features. Polynomial regression is utilized in this study as one of the non-linear methods to evaluate its performance alongside other algorithms. The polynomial regressor was constructed using a polynomial features preprocessor of degree 2 on the training and test set. Afterward, a linear regression model was trained on the generated polynomial features.

2.3.6 Fully Connected Neural Network (FCNN)

A Fully Connected Neural Network (FCNN) is an artificial neural network that contains a series of fully connected layers. The data is passed through the layers, starting from the input layer, then to one or more hidden layers, and finally to the output layer. The connection between neurons has an associated weight, adjusted throughout the training process to minimize the loss or prediction error. The layers contain an activation function applied to each neuron, which introduces non-linearity and allows the network to understand complex patterns in the data. This study uses FCNN as a core deep learning model for prediction. The network architecture contains the input layer that combines the original features with augmented features generated by the machine learning models. It has two (02) hidden layers with 128 neurons, each with the ReLU activation function and an output layer.

2.3.7 Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is a recurrent neural network (RNN) designed to learn and retain information from long data sequences. LSTMs can use neurons called memory cells to retain information for a long time. Each cell of LSTM has three key components: an input gate, a forget gate, and an output gate. These gates maintain the flow of information from inside and outside the cell, allowing the network to remember or forget selective details when needed. LSTM is also used as a core deep learning model for prediction. The inputs provided to LSTM are augmented features generated by the ML models. The LSTM model comprises an input layer; the input shape is the number of sequences, timesteps, and features, followed by an LSTM layer of 64 neurons, the activation function of ReLU, and the ending with one output layer.

2.3.8 Bidirectional Long Short-Term Memory (Bi-LSTM)

Bidirectional Long Short-Term Memory (Bi-LSTM) is an RNN that extends the functionality of LSTM by processing a sequence of input in both directions, forward and backward. For each step, Bi-LSTM has access to both the past and future context, allowing the model to capture the relationship between input sequences. In our study, Bi-LSTM is also used as a core model for prediction. The inputs for this model are the augmented features gathered from the ML models. The Bi-LSTM model's architecture is comprised of an input layer. The shape for the model was several sequences, timesteps, and several features, then connected with a Bi-LSTM layer of 64 neurons, an activation function of ReLU, and ending with one output layer of 01 neurons.

2.4 Model training and evaluation

The training of each model is performed by splitting the dataset in an 80/20 ratio, i.e., 80% for training and 20% for testing. For deep learning models, a validation split of 20% is also used in the training set so that the model can adjust the weights and minimize loss during training. Deep learning models are compiled using the Adam optimizer and Mean Squared Error (MSE) as the loss metric. An early stopping condition of 10 epochs is set to monitor validation loss, with best weights being restored.

2.4.1 Evaluation metrics

In the present study, we are dealing with a regression problem, and each model is evaluated using four (04) metrics, i.e., Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-Squared (R^2).

2.4.1.1 Mean Squared Error (MSE) Mean Squared Error (MSE) measures the average squared difference between the dataset's actual values and the model's predicted values. It is calculated by taking the average squared difference between each actual value and its predicted value. It gives more weight to significant errors because the mistakes in this metric are squared. A lower MSE shows a better fit of the model to the dataset, and a higher MSE shows that the model's predictions are further from true values, indicating a poor performance.

2.4.1.2 Root Mean Squared Error (RMSE) Root Mean Squared Error (RMSE) is the square root of MSE. Since MSE involves squared errors, this metric aims to bring the units of the error back to the same scale as the original data. This metric allows for more interpretation in the context of original data, making it easier to understand the magnitude of the errors.

2.4.1.3 Mean Absolute Error (MAE) Mean Absolute Error (MAE) is a metric used in regression to measure the average absolute difference between the actual dataset values and the predicted model values. It directly represents the average magnitude of errors in the model's prediction. Unlike MSE, it does not square the mistake. Instead, it treats all errors equally and is more robust to outliers than MSE and RMSE.

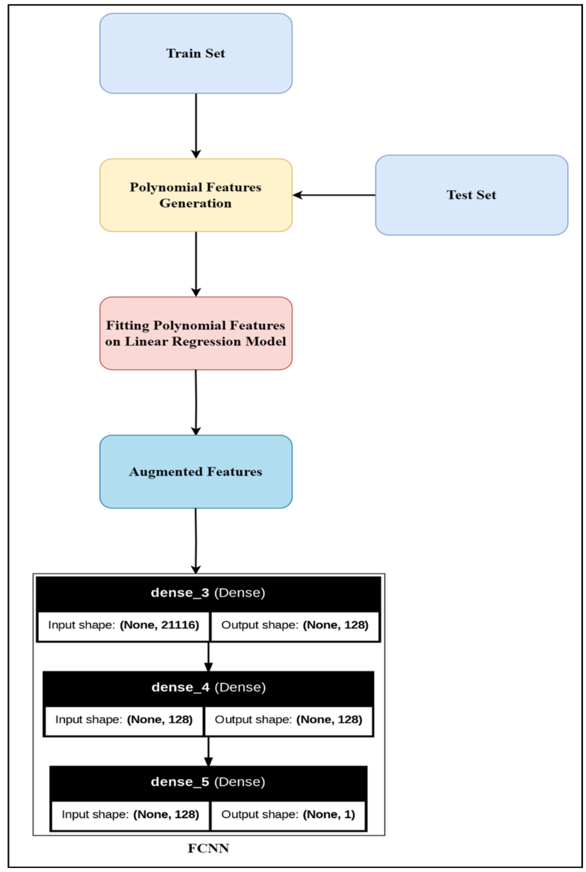

Fig. 8 Architecture of our proposed model for SOGA

2.4.1.4 R-Squared (R2) R-squared, or the coefficient of determination, is a statistical metric that shows how well the model predictions approximate the data points. The value of R-squared ranges from 0 to 1, where '1' indicates that the model perfectly predicts the data and 0 indicates that the model is unable to predict any variability.

In the present study, we have also proposed a hybrid model of PR-FCNN for both the datasets whose architecture is shown in Figs. 8 and 9 respectively.

3 Results

This section provides results from implementing various models to the State of Global Air (SOGA) and World Air Quality Index (WAQI) datasets. The discussion is organized according to this study's evaluation metrics: mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R2). The results of these metrics show a comprehensive evaluation of the model's predictive capabilities concerning PM2.5 levels.

3.1 Mean Squared Error (MSE)

MSE is one of the essential metrics for evaluating the accuracy of predictive models, as it shows the average squared difference between actual and predicted values. MSE of lower values shows better model performance with more minor prediction errors. Table 1 shows the results of the models obtained when trained and evaluated on the SOGA dataset. The Polynomial Regression + FCNN achieved the lowest MSE of 0.000506, indicating the data's predictive accuracy. The model can effectively capture the patterns within the data. Random Forest + LSTM + Bi-LSTM achieved the highest MSE of 0.001729, indicating a lower level of accuracy when compared to other models. Figure 10 shows the graphical representation of the MSE results. For the WAQI dataset, Table 2 provides the results obtained on the WAQI dataset and

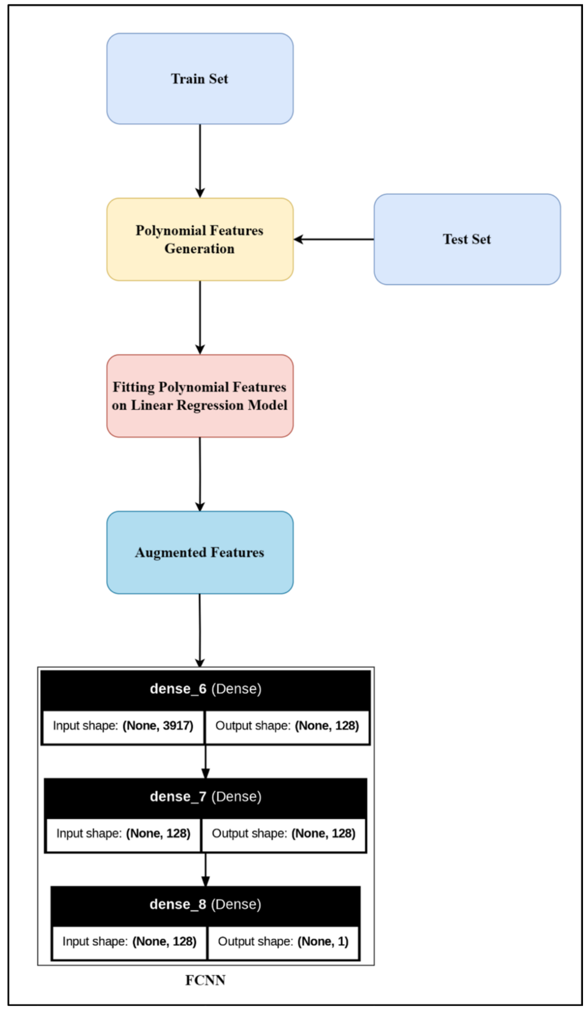

Fig. 9 Architecture of our proposed model for WAQI

Table 1 MSE results on the SOGA dataset

Model	MSE
Polynomial Regression + FCNN	0.0005058
Linear Regression + FCNN	0.000518124
Linear Regression + Bi-LSTM	0.000518482
XGBoost + FCNN	0.000536156
LSTM + Bi-LSTM	0.0005728
FCNN	0.000694718
Polynomial Regression + Bi-LSTM	0.00072085
Extra Trees Regressor + FCNN	0.001258936
Extra Trees Regressor + LSTM + Bi-LSTM	0.001466721
Random Forest + LSTM	0.001644995
Random Forest + FCNN	0.001701687
Random Forest + LSTM + Bi-LSTM	0.001728902

Fig. 11 provides the graphical representation. PR + FCNN achieved the lowest MSE, 0.000261, with ETR + LSTM + Bi-LSTM achieving 0.000267 and RF + LSTM achieving 0.000279. The results on the WAQI dataset indicate that the PR + FCNN

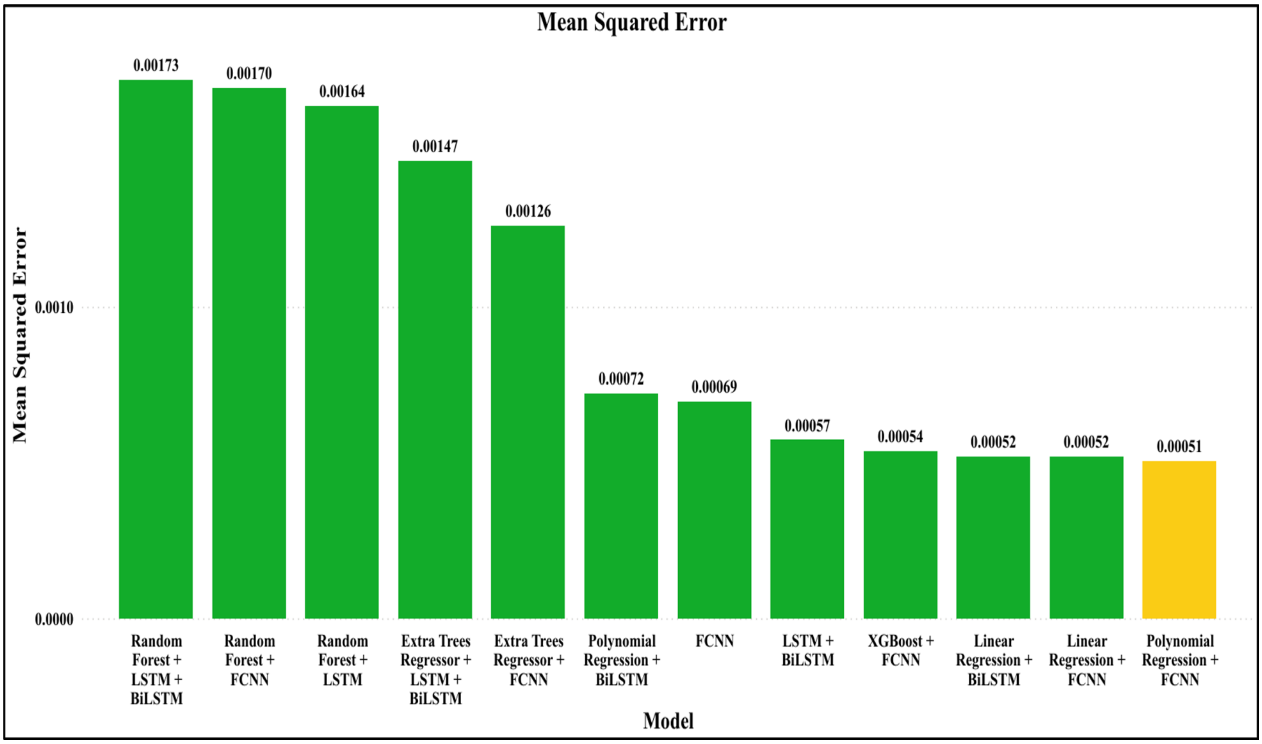

Fig. 10 MSE by models on SOGA dataset

Table 2 MSE results on the WAQI dataset

Model	MSE
Polynomial Regression + FCNN	0.000261102
Extra Trees Regressor + LSTM + Bi-LSTM	0.000266777
Random Forest + LSTM	0.000279166
Random Forest + FCNN	0.000280214
Random Forest + LSTM + Bi-LSTM	0.000299081
XGBoost + FCNN	0.000403908
LSTM + Bi-LSTM	0.00040754
Linear Regression + FCNN	0.000431739
Extra Trees Regressor + FCNN	0.000437044
Linear Regression + Bi-LSTM	0.000451734
Polynomial Regression + Bi-LSTM	0.000451969
FCNN	0.000456372

model is consistent and provides the most negligible squared errors, showing its capability to capture the variance in PM2.5 levels effectively.

3.2 Root Mean Squared Error (RMSE)

RMSE is the square root of MSE, and it provides an error metric that is on the same scale as the original data, which makes it easier to interpret. RMSE values that are lower show better model performance. For the SOGA dataset, Table 3 presents the results on the SOGA dataset and Fig. 12: RMSE by models on SOGA dataset shows a graphical representation of RMSE. Similarly to the MSE results, PR + FCNN performed well and achieved the lowest RMSE of 0.02249, and RF + LSTM + Bi-LSTM achieved the highest RMSE of 0.04158. Table 4 provides the results of RMSE on the WAQI dataset and Fig. 13 shows the graphical representation of RMSE. PR + FCNN achieved the lowest RMSE of 0.016159, followed by

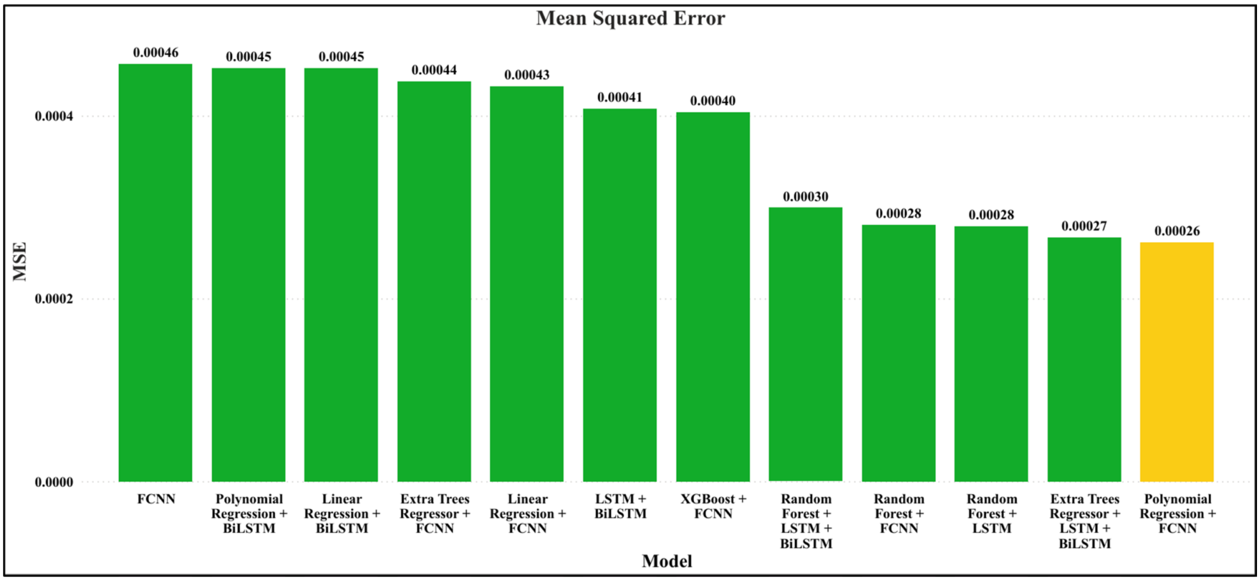

Fig. 11 MSE by models on the WAQI dataset

Table 3 RMSE results on the SOGA dataset

Model	RMSE
Polynomial Regression + FCNN	0.022489995
Linear Regression + FCNN	0.022762345
Linear Regression + Bi-LSTM	0.022770196
XGBoost + FCNN	0.023155035
LSTM + Bi-LSTM	0.02393323
FCNN	0.026357509
Polynomial Regression + Bi-LSTM	0.026848642
Extra Trees Regressor + FCNN	0.035481494
Extra Trees Regressor + LSTM + Bi-LSTM	0.038297791
Random Forest + LSTM	0.040558536
Random Forest + FCNN	0.041251503
Random Forest + LSTM + Bi-LSTM	0.041580074

ETR + LSTM + Bi-LSTM, which had an RMSE of 0.016333, and FCNN, which achieved an RMSE of 0.021362. RMSE values followed the trends observed in MSE, showing the consistency of the model's predictive accuracy.

3.3 Mean Absolute Error (MAE)

MAE presents the average magnitude of errors in the model's predictions, offering an absolute measure of model accuracy. Table 5 and Fig. 14: MAE by models on SOGA dataset. These represent the MAE results of the SOGA dataset. PR + FCNN achieved the lowest MAE, 0.012312, showing that it made the most minor average errors. Similarly to previous results, RF + LSTM + Bi-LSTM achieved the lowest MAE. Table 6 and Fig. 15: MAE by models on the WAQI dataset These represent the MAE results of the WAQI dataset. PR + FCNN achieved the lowest MAE of 0.009892, followed by ETR + LSTM + Bi-LSTM, which got an MAE of 0.009907, and FCNN, which achieved the lowest MAE of 0.013533.

3.4 R-Squared (R2)

R-squared shows how well the model predictions approximate the actual data points, with values closer to 1 indicating better model performance. Table 7, this section presents the R2 results obtained on the SOGA dataset. PR + FCNN performed best and achieved the highest R2 of 0.980861, which shows that it approximates 98% of the variance in PM2.5

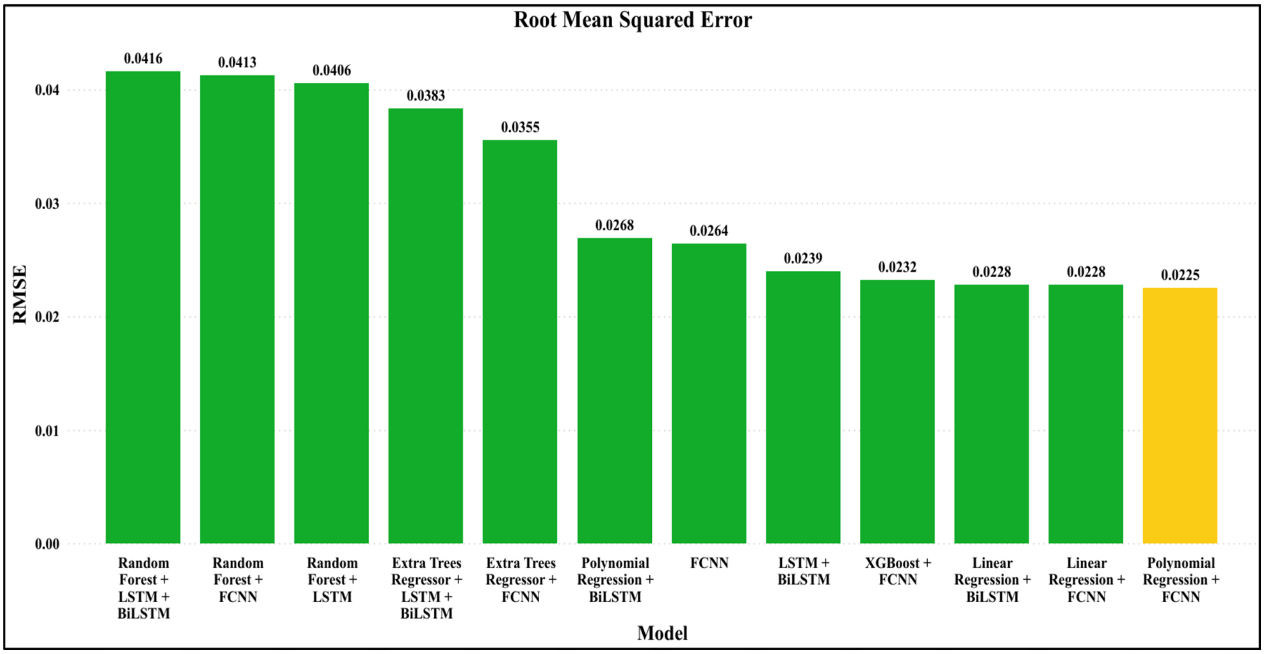

Fig. 12 RMSE by models on SOGA dataset

Table 4 RMSE results on WAQI dataset

Model	RMSE
Polynomial Regression + FCNN	0.016158661
Extra Trees Regressor + LSTM + Bi-LSTM	0.016333317
Random Forest + LSTM	0.016708257
Random Forest + FCNN	0.016739605
Random Forest + LSTM + Bi-LSTM	0.017293969
XGBoost + FCNN	0.020097455
LSTM + Bi-LSTM	0.02018762
Linear Regression + FCNN	0.02077833
Extra Trees Regressor + FCNN	0.020905599
Linear Regression + Bi-LSTM	0.021254025
Polynomial Regression + Bi-LSTM	0.021259556
FCNN	0.021362871

levels. It was followed by LR + FCNN and LR + Bi-LSTM, which achieved R2 of 0.980394 and 0.980380 respectively. Figure 16 presents the graphical representation of R2. Results of R2 obtained on the WAQI dataset are presented in Table 8 and Fig. 17 shows the graphical representation. Similarly to the R2 results of the SOGA dataset, PR + FCNN performed best and achieved an R2 of 0.837154, followed by ETR + LSTM + Bi-LSTM, which attained an R-squared of 0.833614. FCNN obtained the lowest R2 value of 0.715367. Overall, R2 values were lower than those in the SOGA dataset due to differing data quality or variability in the datasets.

As most studies have only considered R^2 as the evaluation metric, we have compared the performance of the state-of-the-art models with the evaluation metric R-squared. The results are shown in Table 9 indicate the optimized performance of our model.

From the results of all metrics, the Polynomial Regression + FCNN model was consistent in its performance, showing its ability to capture linear and non-linear patterns in the data effectively. The low MSE, RMSE, and MAE values, with high R2 values, present that our proposed PR + FCNN model can accurately predict PM2.5 levels with minimal errors, followed by other models such as LR + FCNN and ETR + LSTM + Bi-LSTM also provided consistent performance across different metrics, showing the effectiveness of hybrid modeling approach. Overall, this study explored the efficacy of hybrid modeling

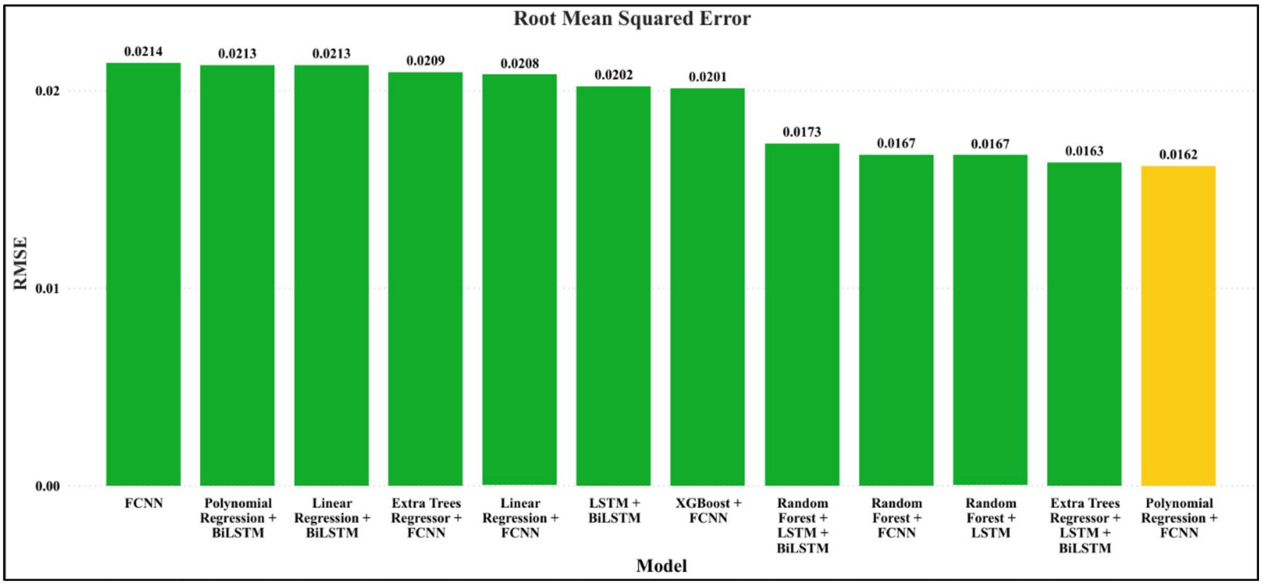

Fig. 13 RMSE by models on the WAQI dataset

Table 5 MAE results on the SOGA dataset

Model	MAE
Polynomial Regression + FCNN	0.012312043
Linear Regression + FCNN	0.013045439
Linear Regression + Bi-LSTM	0.012580553
XGBoost + FCNN	0.015059124
LSTM + Bi-LSTM	0.013639153
FCNN	0.016020088
Polynomial Regression + Bi-LSTM	0.015579839
Extra Trees Regressor + FCNN	0.015473997
Extra Trees Regressor + LSTM + Bi-LSTM	0.014584173
Random Forest + LSTM	0.022914739
Random Forest + FCNN	0.02264623
Random Forest + LSTM + Bi-LSTM	0.023357216

approaches in air quality prediction. Our proposed model PR + FCNN was the best performer due to its ability to capture complicated relationships in the data. Further improvement of the model and exploration of model combinations may result in even more reliable and precise predictions in the future.

4 Conclusion

This research demonstrated the effectiveness of hybrid models combining machine learning and deep learning techniques in predicting PM2.5 concentrations. Among various models tested, the Polynomial Regression (PR) combined with Fully Connected Neural Networks (FCNN) consistently outperformed other models, showing the lowest Mean Squared Error (MSE) and the highest R-squared (R^2) value across different datasets. The model's ability to capture linear and non-linear relationships in the data contributed to its superior predictive performance. The combination of machine learning

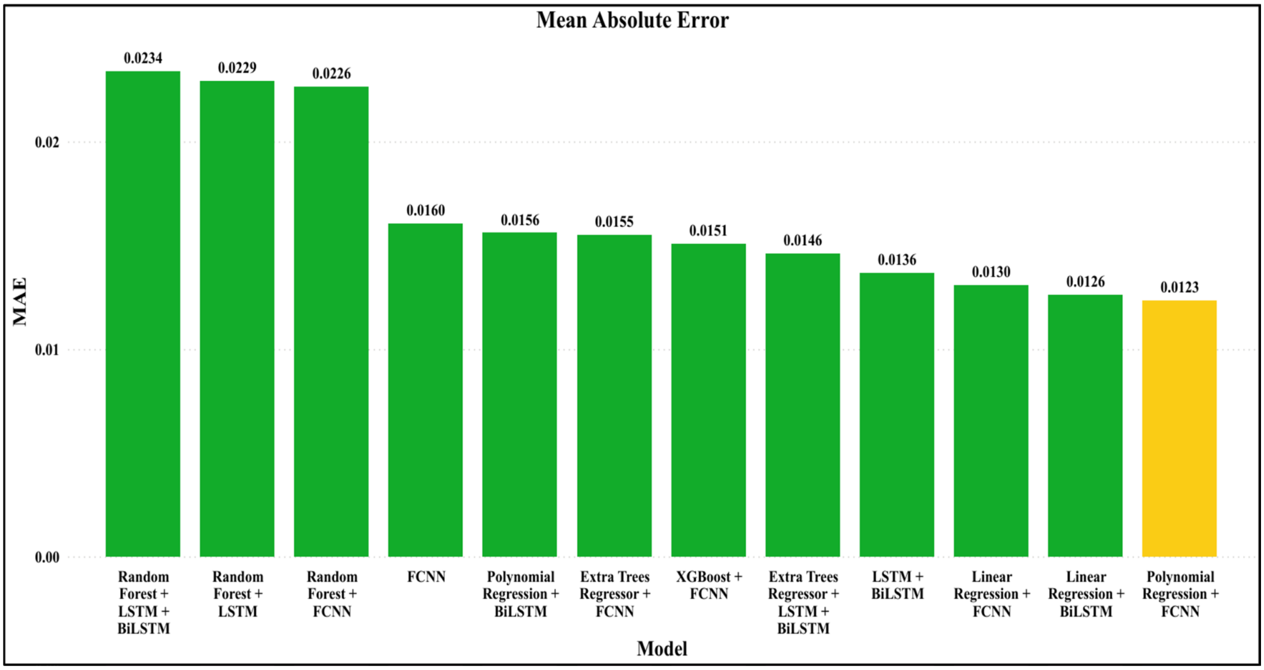

Fig. 14 MAE by models on SOGA dataset

Table 6 MAE results on the WAQI dataset

Model	MAE
Polynomial Regression + FCNN	0.009892059
Extra Trees Regressor + LSTM + Bi-LSTM	0.009907342
Random Forest + LSTM	0.010432687
Random Forest + FCNN	0.010396563
Random Forest + LSTM + Bi-LSTM	0.010770933
XGBoost + FCNN	0.013071065
LSTM + Bi-LSTM	0.013184642
Linear Regression + FCNN	0.013309105
Extra Trees Regressor + FCNN	0.013276778
Linear Regression + Bi-LSTM	0.01356522
Polynomial Regression + Bi-LSTM	0.013557477
FCNN	0.013533644

models, such as Random Forest (RF), XGBoost, and Extra Trees Regressor (ETR), with deep learning models like LSTM and Bi-LSTM further enhanced the model's capacity to handle complex temporal and spatial data, resulting in more accurate predictions. The study also underscores the importance of integrating multiple datasets, such as those from the World Air Quality Index (WAQI) and the State of Global Air, for robust modeling. The success of the hybrid models in capturing PM2.5 patterns emphasizes the potential for applying similar approaches to other air pollutants and broader environmental issues. In conclusion, this research highlights the potential of hybrid modeling approaches to significantly improve air quality predictions, which could play a crucial role in mitigating the health impacts of air pollution. Future studies could focus on refining these models and incorporating additional meteorological and socio-economic factors to enhance prediction accuracy further. The findings contribute valuable insights into air quality forecasting, offering practical applications in public health planning and environmental management.

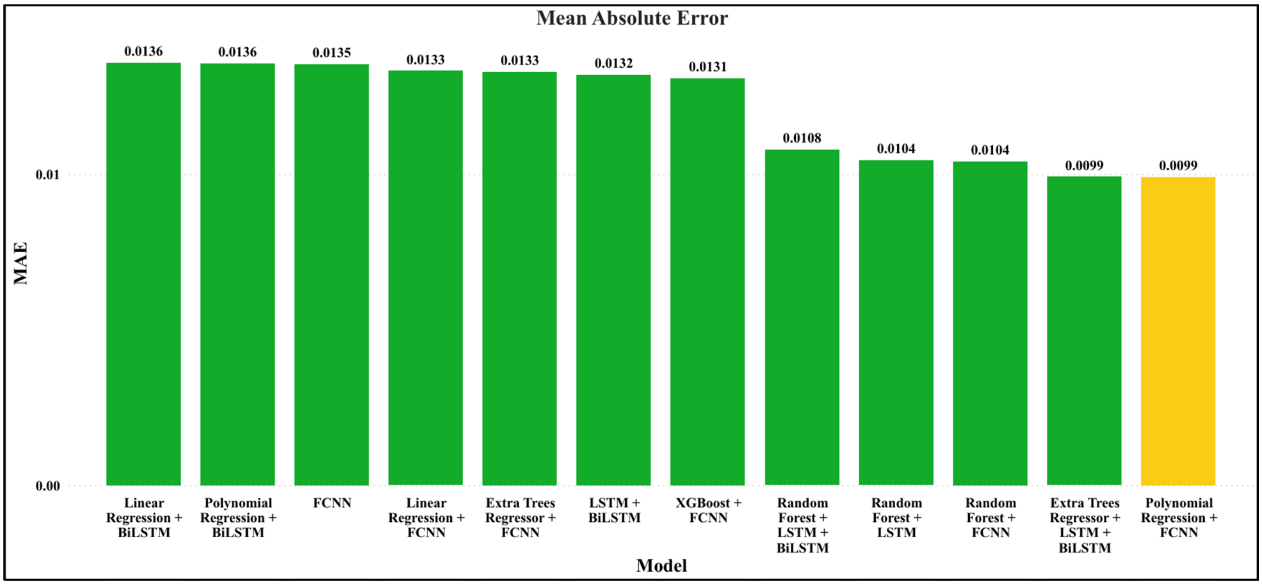

Fig. 15 MAE by models on the WAQI dataset

Table 7 R2 results on the SOGA dataset

Model	R^2
Polynomial Regression + FCNN	0.980860651
Linear Regression + FCNN	0.980394304
Linear Regression + Bi-LSTM	0.980380774
XGBoost + FCNN	0.979712009
LSTM + Bi-LSTM	0.978325427
FCNN	0.973712027
Polynomial Regression + Bi-LSTM	0.972723246
Extra Trees Regressor + FCNN	0.95236212
Extra Trees Regressor + LSTM + Bi-LSTM	0.944499612
Random Forest + LSTM	0.937753797
Random Forest + FCNN	0.935608625
Random Forest + LSTM + Bi-LSTM	0.934578776

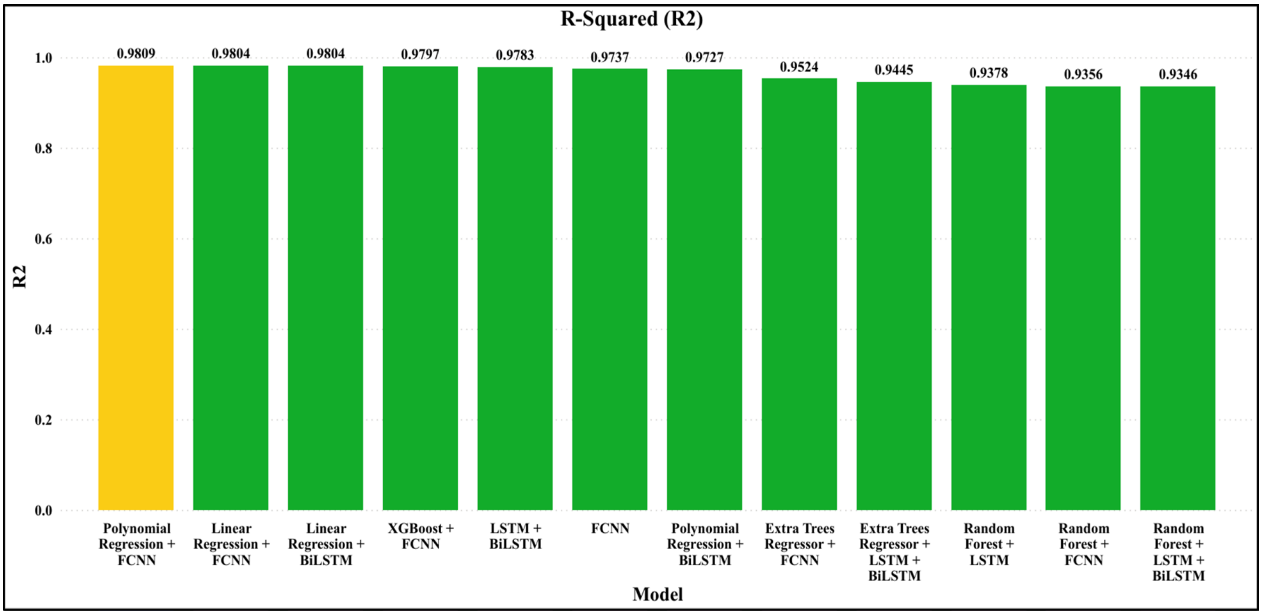

Fig. 16 R2 by models on the SOGA dataset

Table 8 R2 results on the WAQI dataset

Model	R^2
Polynomial Regression + FCNN	0.83715421
Extra Trees Regressor + LSTM + Bi-LSTM	0.833614886
Random Forest + LSTM	0.825888276
Random Forest + FCNN	0.825234294
Random Forest + LSTM + Bi-LSTM	0.813467264
XGBoost + FCNN	0.748088539
LSTM + Bi-LSTM	0.745823145
Linear Regression + FCNN	0.730730534
Extra Trees Regressor + FCNN	0.72742188
Linear Regression + Bi-LSTM	0.718260229
Polynomial Regression + Bi-LSTM	0.718113601
FCNN	0.715367138

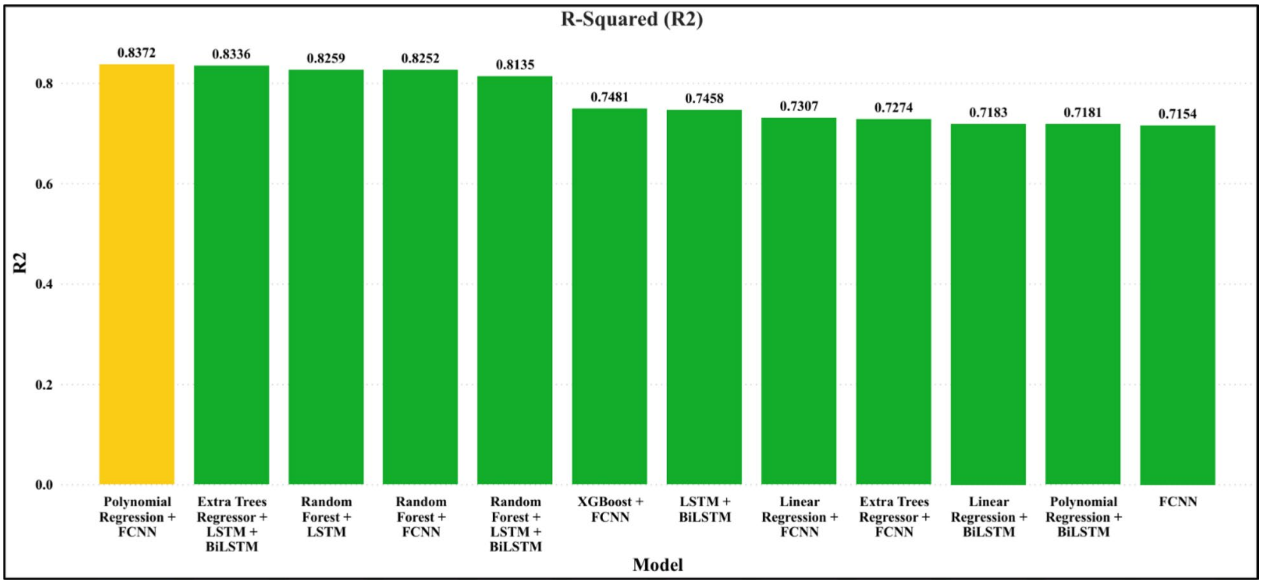

Fig. 17 R2 by models on the WAQI dataset

Table 9 Comparison of our proposed model with state-of-the-art

S. No	Model	Year	R^2
1	XGBoost [46]	2019	0.8100
2	DBN [50]	2019	0.8840
3	LSTM [38]	2021	0.2920
4	TDBN [49]	2021	0.8620
5	M^2-APNET [19]	2023	0.6000
6	AdaBoost-LSTM [48]	2024	0.9007
7	PR-FCNN (our proposed model)	2024	0.9809

Author contributions All authors have equally contributed.

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Data availability The datasets analyzed during the current study are available in the https://github.com/syedazeeminam/performance_analysis_pm2.5 repository and can also be found on the following link given below: https://www.stateofglobalair.org/data/#/air/table?pollutant=pm25; https://aqicn.org/data-platform/covid19/.

Declarations

Ethics approval and consent to participate The work does not involve human participants, human data, or human tissue.

Consent for publication The work contains no person's data.

Competing interests The authors declare no competing interests.

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